

# A COUNTEREXAMPLE TO MAKEEV'S REGULAR-SIMPLEX CONJECTURE IN DIMENSION FOUR

WORKING DRAFT

ABSTRACT. Let  $u_{ij} = (e_i - e_j)/\sqrt{2}$  be the normalized roots of type  $A_4$ . We give an explicit smooth odd function on  $S^3$  which does not agree with any linear functional on any rotated copy of these roots. After identifying  $\mathbb{R}^4 \cong \mathbb{C}^2$ , for every sufficiently small nonzero real  $\varepsilon$  one may take

$$f_\varepsilon(z_1, z_2) = \operatorname{Re}(z_1^2 \bar{z}_2) + \varepsilon |z_1|^2 \operatorname{Im}(z_1^2 \bar{z}_2).$$

The proof uses a cyclic model of the regular 4-simplex, the exact-label decomposition of edge flows, a cubic identity forced by a circle symmetry, and an explicit algebraic transversality calculation.

## 1. INTRODUCTION

Kuperberg showed that a constant-width formulation of Makeev's universal-cover conjecture is equivalent to asking whether every continuous odd function on  $S^{d-1}$  agrees with a linear function on a rotated  $A_d$  root system. He proved the assertion for  $d = 3$  and showed that the homological obstruction used there vanishes in higher dimensions [2]. The result below gives a negative answer in the first higher dimension.

The roots must be normalized because the function is defined on the unit sphere. Thus the standard model is

$$V = \mathbf{1}^\perp \subset \mathbb{R}^5, \quad u_{ij} = \frac{e_i - e_j}{\sqrt{2}} \in S(V), \quad 0 \leq i < j \leq 4.$$

The configuration has an orientation-reversing symmetry, induced for example by a transposition of two coordinate vectors. Consequently the formulations using  $O(4)$  and  $SO(4)$  are equivalent.

**Theorem 1.1** (Counterexample). *There exists  $\varepsilon_0 > 0$  such that, for every real  $\varepsilon$  with  $0 < |\varepsilon| < \varepsilon_0$ , the smooth odd function*

$$f_\varepsilon(z_1, z_2) = \operatorname{Re}(z_1^2 \bar{z}_2) + \varepsilon |z_1|^2 \operatorname{Im}(z_1^2 \bar{z}_2), \quad (z_1, z_2) \in S^3, \quad (1)$$

*has no  $A \in SO(4)$  and  $b \in \mathbb{R}^4$  satisfying*

$$f_\varepsilon(Au_{ij}) = b \cdot Au_{ij} \quad (0 \leq i < j \leq 4).$$

## 2. THE CYCLIC SIMPLEX AND EXACT EDGE LABELS

Identify  $\mathbb{R}^4$  with  $\mathbb{C}^2$ , equipped with the real inner product

$$\langle z, w \rangle_{\mathbb{R}} = \operatorname{Re}(z_1 \bar{w}_1 + z_2 \bar{w}_2).$$

Let  $\zeta = e^{2\pi i/5}$  and put

$$q_j = \sqrt{\frac{2}{5}}(\zeta^j, \zeta^{2j}), \quad j = 0, \dots, 4. \quad (2)$$

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This manuscript reorganizes and simplifies the supplied argument. It makes no claim of priority.

**Lemma 2.1** (Cyclic regular simplex). *The vectors  $q_0, \dots, q_4$  satisfy*

$$\sum_{j=0}^4 q_j = 0, \quad \langle q_i, q_j \rangle_{\mathbb{R}} = \delta_{ij} - \frac{1}{5}.$$

*Consequently*

$$u_{ij} = \frac{q_i - q_j}{\sqrt{2}} \quad (3)$$

*are unit vectors forming an  $A_4$  root system. This system is orientation-preservingly isometric to the normalized standard system.*

*Proof.* Both  $\sum_j \zeta^j$  and  $\sum_j \zeta^{2j}$  vanish. Moreover,

$$\langle q_i, q_j \rangle_{\mathbb{R}} = \frac{2}{5} \operatorname{Re}(\zeta^{i-j} + \zeta^{2(i-j)}).$$

For  $i \neq j$  the real part is  $-1/2$ , whereas for  $i = j$  it is 2. Thus the displayed Gram matrix holds and  $\|q_i - q_j\|^2 = 2$ . The map  $e_j - \frac{1}{5}\mathbf{1} \mapsto q_j$  is therefore an isometry from the standard simplex. Composing it, if necessary, with an odd permutation of the vertices makes it orientation-preserving without changing the unoriented root system.  $\square$

By the last assertion, it is enough to prove [theorem 1.1](#) for the cyclic roots (3); we use the same notation  $u_{ij}$  for them from now on.

For an odd function  $F: S^3 \rightarrow \mathbb{R}$  and  $g \in \operatorname{SO}(4)$ , define the skew matrix

$$Y_F(g)_{ij} = F(gu_{ij}), \quad Y_F(g)_{ji} = -Y_F(g)_{ij}. \quad (4)$$

Let

$$\Pi = I - \frac{1}{5}\mathbf{1}\mathbf{1}^T.$$

**Lemma 2.2** (Exact edge labels). *For a real skew  $5 \times 5$  matrix  $Y$ , the following are equivalent:*

- (1) *there exists  $c \in \mathbb{R}^4$  such that  $Y_{ij} = c \cdot u_{ij}$ ;*
- (2) *there exist real numbers  $\rho_0, \dots, \rho_4$  such that  $Y_{ij} = \rho_i - \rho_j$ ;*
- (3)  $\Pi Y \Pi = 0$ .

*Proof.* If  $Y_{ij} = c \cdot u_{ij}$ , take  $\rho_i = (c \cdot q_i)/\sqrt{2}$ . Conversely, subtract the mean of the  $\rho_i$  and put

$$c = \sqrt{2} \sum_i \rho_i q_i.$$

The simplex Gram matrix gives  $c \cdot q_j = \sqrt{2}\rho_j$ , hence  $c \cdot u_{ij} = \rho_i - \rho_j$ .

If  $Y_{ij} = \rho_i - \rho_j$ , then  $Y = \rho\mathbf{1}^T - \mathbf{1}\rho^T$ , so  $\Pi Y \Pi = 0$ . Conversely, write  $J_0 = \frac{1}{5}\mathbf{1}\mathbf{1}^T$ . Since  $Y$  is skew,  $J_0 Y J_0 = 0$ , and  $\Pi Y \Pi = 0$  gives  $Y = J_0 Y + Y J_0$ . With  $\rho_i = \frac{1}{5} \sum_j Y_{ij}$ , this identity reads  $Y_{ij} = \rho_i - \rho_j$ .  $\square$

This is the complete-graph cut/cycle decomposition, or equivalently the one-dimensional part of combinatorial Hodge theory for edge flows [1]. We call such a matrix *exact*. The equality in [theorem 1.1](#) holds at a rotation  $g$  precisely when  $Y_F(g)$  is exact: if  $F(gu_{ij}) = b \cdot gu_{ij}$ , take  $c = g^{-1}b$ , and conversely take  $b = gc$ .

We shall also use the elementary consequence that every skew matrix  $X$  with  $X\mathbf{1} = 0$  annihilates exact labels:

$$\sum_{i < j} X_{ij}(\rho_i - \rho_j) = 0. \quad (5)$$

## 3. A SYMMETRY FUNCTIONAL FOR THE CUBIC

Put

$$W(z) = z_1^2 \bar{z}_2, \quad Q = \operatorname{Re} W, \quad R = \operatorname{Im} W, \quad H = |z_1|^2 R. \quad (6)$$

Consider the circle action

$$h_\theta(z_1, z_2) = (e^{i\theta} z_1, e^{2i\theta} z_2), \quad J(z_1, z_2) = (iz_1, 2iz_2). \quad (7)$$

The functions  $Q$ ,  $R$ , and  $H$  are invariant under  $h_\theta$ .

For  $g \in \operatorname{SO}(4)$  write  $p_i = gq_i$  and define

$$X(g)_{ij} = \langle p_i, Jp_j \rangle_{\mathbb{R}}, \quad \Lambda_g(Y) = \sum_{i < j} X(g)_{ij} Y_{ij}. \quad (8)$$

The matrix  $X(g)$  is skew and

$$\sum_j X(g)_{ij} = \left\langle p_i, J \sum_j p_j \right\rangle_{\mathbb{R}} = 0.$$

Thus (5) shows that  $\Lambda_g$  annihilates every exact label matrix.

The following cubic identity is the only remnant of the variational cubic argument needed for the counterexample. Let

$$P_3(x) = \sum_{i=0}^4 \langle q_i, x \rangle_{\mathbb{R}}^3.$$

For a homogeneous cubic  $C$ , write  $T_C$  for its symmetric trilinear polarization. For any orthonormal basis  $e_1, \dots, e_4$ , set

$$\langle C, D \rangle = \sum_{\alpha, \beta, \gamma=1}^4 T_C(e_\alpha, e_\beta, e_\gamma) T_D(e_\alpha, e_\beta, e_\gamma).$$

This is independent of the chosen orthonormal basis. For  $K \in \mathfrak{so}(4)$ , let  $(K \cdot C)(x) = \left. \frac{d}{dt} \right|_{t=0} C(e^{-tK} x)$ .

**Lemma 3.1** (Cubic edge identity). *For every homogeneous cubic  $C$  and  $K \in \mathfrak{so}(4)$ ,*

$$\langle C, K \cdot P_3 \rangle = 2\sqrt{2} \sum_{i < j} \langle q_i, Kq_j \rangle_{\mathbb{R}} C(u_{ij}). \quad (9)$$

*Proof.* The simplex is a Parseval frame:  $\sum_i q_i q_i^T = I$ . Hence

$$\langle C, K \cdot P_3 \rangle = 3 \sum_i T_C(q_i, q_i, Kq_i).$$

On the other hand, because  $C(u_{ij}) = C(q_i - q_j)/(2\sqrt{2})$ , the right side of (9) equals

$$\sum_{i < j} \langle q_i, Kq_j \rangle_{\mathbb{R}} C(q_i - q_j).$$

Expanding the cubic, the pure terms vanish since  $\sum_j \langle q_i, Kq_j \rangle_{\mathbb{R}} = 0$ . The two mixed terms combine to

$$-3 \sum_{i,j} \langle q_i, Kq_j \rangle_{\mathbb{R}} T_C(q_i, q_i, q_j).$$

The Parseval identity gives  $Kq_i = -\sum_j \langle q_i, Kq_j \rangle_{\mathbb{R}} q_j$ , so the last expression is  $3 \sum_i T_C(q_i, q_i, Kq_i)$ , as required.  $\square$

**Lemma 3.2** (Global cubic annihilation). *For every  $g \in \operatorname{SO}(4)$ ,*

$$\Lambda_g(Y_Q(g)) = 0. \quad (10)$$

*Proof.* Take  $K = g^{-1}Jg$  and  $C = Q \circ g$  in Lemma 3.1. Then

$$\langle q_i, Kq_j \rangle_{\mathbb{R}} = X(g)_{ij}.$$

The orthogonal action on cubic tensors is infinitesimally skew-adjoint, while  $K \cdot (Q \circ g) = (J \cdot Q) \circ g = 0$  by the circle invariance of  $Q$ . Therefore

$$\langle Q \circ g, K \cdot P_3 \rangle = -\langle K \cdot (Q \circ g), P_3 \rangle = 0,$$

and (9) gives (10).  $\square$

The remaining ingredient is an explicit algebraic statement. Its proof is contained in section A.

**Proposition 3.3** (Algebraic transversality). *If  $Y_Q(g)$  is exact, then*

$$\Lambda_g(Y_H(g)) \neq 0. \quad (11)$$

#### 4. COMPLETION OF THE COUNTEREXAMPLE

We now prove theorem 1.1. If the conclusion were false, there would be nonzero numbers  $\varepsilon_\nu \rightarrow 0$  and rotations  $g_\nu \in \text{SO}(4)$  for which the labels of  $Q + \varepsilon_\nu H$  are exact. By Lemma 2.2,

$$\Pi Y_Q(g_\nu) \Pi + \varepsilon_\nu \Pi Y_H(g_\nu) \Pi = 0. \quad (12)$$

After passing to a subsequence, compactness gives  $g_\nu \rightarrow g$ . The second term in (12) tends to zero, so  $\Pi Y_Q(g) \Pi = 0$ ; hence  $Y_Q(g)$  is exact.

Because  $\Lambda_{g_\nu}$  annihilates exact labels and  $\Lambda_{g_\nu}(Y_Q(g_\nu)) = 0$  by Lemma 3.2, exactness of  $Y_Q(g_\nu) + \varepsilon_\nu Y_H(g_\nu)$  gives

$$\Lambda_{g_\nu}(Y_H(g_\nu)) = 0.$$

Passing to the limit yields  $\Lambda_g(Y_H(g)) = 0$ . This contradicts Proposition 3.3, because  $Y_Q(g)$  is exact. Therefore an  $\varepsilon_0 > 0$  exists, and every function in (1) with  $0 < |\varepsilon| < \varepsilon_0$  is a counterexample. This proves theorem 1.1.

#### APPENDIX A. PROOF OF ALGEBRAIC TRANSVERSALITY

We prove Proposition 3.3. Fix  $g \in \text{SO}(4)$  for which the  $Q$ -labels are exact, write

$$p_i = gq_i = (a_i, b_i) \in \mathbb{C}^2, \quad a = (a_0, \dots, a_4), \quad b = (b_0, \dots, b_4) \in \mathbb{C}^5,$$

and use coordinatewise multiplication of vectors in  $\mathbb{C}^5$ . The Hermitian inner product is conjugate-linear in the first variable:

$$\langle x, y \rangle_{\mathbb{C}^5} = \sum_i \bar{x}_i y_i.$$

**A.1. Frame identities.** The  $5 \times 4$  real matrix whose rows are the  $p_i$  has orthonormal columns, all perpendicular to  $\mathbf{1}$ . Therefore

$$\sum_i a_i = \sum_i b_i = 0, \quad (13)$$

$$\sum_i a_i^2 = \sum_i b_i^2 = \sum_i a_i b_i = \sum_i a_i \bar{b}_i = 0, \quad (14)$$

$$\sum_i |a_i|^2 = \sum_i |b_i|^2 = 2, \quad (15)$$

$$|a_i|^2 + |b_i|^2 = \frac{4}{5}. \quad (16)$$

Consequently

$$\mathbf{1}, a, b, \bar{a}, \bar{b}$$

are mutually orthogonal in  $\mathbb{C}^5$ , with squared norms 5, 2, 2, 2, 2. Every  $x \in \mathbb{C}^5$  has the expansion

$$x = \frac{\sum_i x_i}{5} \mathbf{1} + \frac{1}{2} \left( \langle a, x \rangle a + \langle b, x \rangle b + \langle \bar{a}, x \rangle \bar{a} + \langle \bar{b}, x \rangle \bar{b} \right). \quad (17)$$

For  $u, v \in \mathbb{C}^5$ , write

$$(u \wedge v)_{ij} = u_i v_j - v_i u_j.$$

**A.2. Cycle equations for the cubic.** At the root  $gu_{ij} = (p_i - p_j)/\sqrt{2}$ ,

$$W(gu_{ij}) = \frac{1}{2\sqrt{2}} (a_i - a_j)^2 (\bar{b}_i - \bar{b}_j).$$

Ignoring the harmless factor  $1/(2\sqrt{2})$ , let

$$\tilde{Y}_{ij} = (a_i - a_j)^2 (\bar{b}_i - \bar{b}_j).$$

Modulo exact matrices, expansion gives

$$\Pi \tilde{Y} \Pi \equiv -a^2 \wedge \bar{b} + 2a \wedge (a\bar{b}). \quad (18)$$

Indeed, the omitted term is  $a_i^2 \bar{b}_i - a_j^2 \bar{b}_j$ .

Define

$$s = \frac{1}{2} \sum_i |a_i|^2 a_i, \quad \lambda = \frac{1}{2} \sum_i |a_i|^2 \bar{b}_i, \quad (19)$$

$$m = \sum_i a_i^2 \bar{b}_i, \quad n = \sum_i a_i^3, \quad u = \frac{1}{2} \sum_i a_i \bar{b}_i^2, \quad \ell = \frac{1}{2} \sum_i a_i^2 b_i. \quad (20)$$

Using (17),

$$a^2 = sa + \frac{m}{2} b + \frac{n}{2} \bar{a} + \ell \bar{b}, \quad (21)$$

$$a\bar{b} = \lambda a + ub + \frac{m}{2} \bar{a} - s\bar{b}. \quad (22)$$

Substitution into (18) yields

$$\omega := \Pi \tilde{Y} \Pi = 2u a \wedge b + m a \wedge \bar{a} - 3s a \wedge \bar{b} - \frac{m}{2} b \wedge \bar{b} - \frac{n}{2} \bar{a} \wedge \bar{b}. \quad (23)$$

Exactness of the real  $Q$ -labels is precisely  $\text{Re } \omega = 0$ . The six wedges of the complex basis  $a, b, \bar{a}, \bar{b}$  are linearly independent. Comparing (23) with its conjugate gives

$$s = 0, \quad m \in \mathbb{R}, \quad 4u = \bar{n}. \quad (24)$$

Replacing  $g$  by  $h_\theta g$  changes neither the values of  $Q$  and  $H$  nor the matrix  $X(g)$ , while  $\lambda \mapsto e^{-2i\theta} \lambda$ . We may therefore assume

$$\lambda = t \geq 0. \quad (25)$$

In addition define

$$v = \sum_i a_i b_i^2, \quad k = \sum_i b_i^3. \quad (26)$$

**A.3. The coordinatewise multiplication table.** Expanding each product in the orthogonal basis  $\mathbf{1}, a, b, \bar{a}, \bar{b}$  by (17), and using (16) and (24), gives

$$\begin{aligned}
a\bar{a} &= \frac{2}{5}\mathbf{1} + tb + t\bar{b}, & b\bar{b} &= \frac{2}{5}\mathbf{1} - tb - t\bar{b}, \\
a^2 &= \frac{m}{2}b + \frac{n}{2}\bar{a} + \ell\bar{b}, & \bar{a}^2 &= \frac{m}{2}\bar{b} + \frac{\bar{n}}{2}a + \bar{\ell}b, \\
a\bar{b} &= ta + \frac{\bar{n}}{4}b + \frac{m}{2}\bar{a}, & \bar{a}b &= t\bar{a} + \frac{n}{4}\bar{b} + \frac{m}{2}a, \\
ab &= ta + \ell\bar{a} + \frac{v}{2}\bar{b}, & \bar{a}\bar{b} &= t\bar{a} + \bar{\ell}a + \frac{\bar{v}}{2}b, \\
b^2 &= \frac{n}{4}a - tb + \frac{v}{2}\bar{a} + \frac{k}{2}\bar{b}, & \bar{b}^2 &= \frac{\bar{n}}{4}\bar{a} - t\bar{b} + \frac{\bar{v}}{2}a + \frac{\bar{k}}{2}b.
\end{aligned} \tag{27}$$

For example, the coefficient of  $b$  in  $a\bar{a}$  is  $\frac{1}{2}\sum_i \bar{b}_i |a_i|^2 = t$ , and the other coefficients are obtained in the same way.

Coordinatewise multiplication is associative. Comparing coefficients in (27) gives the following necessary equations:

$$3mn = 4tv, \tag{E1}$$

$$\bar{n}v + 16\ell t + 8mt = 0, \tag{E2}$$

$$-2km + 24\ell t - n^2 = 0, \tag{E3}$$

$$-kt + \ell m + 4t^2 = 0, \tag{E4}$$

$$\bar{n}k + mn + 8tv = 0, \tag{E5}$$

$$20|\ell|^2 + 5m^2 + 5|n|^2 - 40t^2 = 8, \tag{E6}$$

$$16|\ell|^2 - 4m^2 - |n|^2 + 4|v|^2 = 0, \tag{E7}$$

$$20|\ell|^2 + 60t^2 + 5|v|^2 = 8. \tag{E8}$$

More precisely, (E1)–(E3) are the  $a, b, \bar{b}$  coefficients of  $(a^2)b = a(ab)$ ; (E4)–(E5) are the  $a, b$  coefficients of  $(ab)b = a(b^2)$ ; and (E6), (E7), (E8) are respectively the indicated coefficients of

$$(a^2)\bar{a} = a(a\bar{a}), \quad (ab)\bar{a} = a(b\bar{a}), \quad (ab)\bar{b} = a(b\bar{b}).$$

**A.4. Classification of the possible cubic zeros.** We extract the consequences of (E1)–(E8) needed below.

**Case I:**  $t = 0$ . Equation (E1) gives  $mn = 0$ . If  $m = 0$ , then (E3) gives  $n = 0$ , while (E6)–(E7) become

$$20|\ell|^2 = 8, \quad 16|\ell|^2 + 4|v|^2 = 0,$$

a contradiction. Hence  $m \neq 0$  and  $n = 0$ . Equations (E4) and (E3) give  $\ell = k = 0$ , and then

$$m^2 = \frac{8}{5}, \quad |v|^2 = m^2. \tag{28}$$

**Case II:**  $t > 0$  and  $n = 0$ . Equation (E1) gives  $v = 0$ , and (E2) gives

$$\ell = -\frac{m}{2}. \tag{29}$$

Here  $m \neq 0$ , since otherwise (27) would give  $a^2 = 0$ . Equations (E3) and (E4) give  $k = -6t$  and  $m^2 = 20t^2$ . Finally (E8) yields

$$t^2 = \frac{1}{20}, \quad m^2 = 1. \tag{30}$$

**Case III:**  $t > 0$  and  $n \neq 0$ . First  $m \neq 0$ , since otherwise (E1), (E2), and (E3) successively give  $v = 0$ ,  $\ell = 0$ , and  $n = 0$ . Put

$$q = \frac{m}{t}, \quad h = \frac{|n|^2}{t^2} > 0. \quad (31)$$

Equations (E1) and (E2) give

$$v = \frac{3mn}{4t}, \quad \ell = -\frac{m(32 + 3h)}{64}. \quad (32)$$

In particular,  $\ell$  is real. Equation (E5) gives

$$k = -7m \frac{n}{\bar{n}}.$$

Equation (E4) shows that  $k$  is real, so

$$\sigma := \frac{n}{\bar{n}} \in \{1, -1\}, \quad k = -7m\sigma, \quad n^2 = \sigma |n|^2. \quad (33)$$

After division by appropriate powers of  $t$ , equations (E4) and (E3) become

$$3hq^2 + 32q^2 - 448\sigma q - 256 = 0, \quad (34)$$

$$-9hq - 8\sigma h + 112\sigma q^2 - 96q = 0. \quad (35)$$

Eliminating  $h$  gives

$$(q - 4)(q + 2)(21q^2 + 42q + 16) = 0 \quad (\sigma = 1), \quad (36)$$

$$(q + 4)(q - 2)(21q^2 - 42q + 16) = 0 \quad (\sigma = -1). \quad (37)$$

The roots  $q = -2\sigma$  give  $h = -64$  and are impossible. One root of each quadratic gives  $h = (-74 - 14\sqrt{105})/3 < 0$ . The candidates  $q = 4\sigma$  give  $h = 32$ ; then (32) and (E1) imply

$$\frac{|\ell|^2}{t^2} = 64, \quad \frac{|v|^2}{t^2} = 288,$$

and the left side of (E7), divided by  $t^2$ , is 2080, a contradiction. The only remaining possibility is

$$q = -\sigma \left(1 - \frac{\sqrt{105}}{21}\right), \quad h = \frac{-74 + 14\sqrt{105}}{3}. \quad (38)$$

Equation (E6) then gives

$$t^2 = \frac{7(5\sqrt{105} - 33)}{1440}, \quad m^2 = \frac{49\sqrt{105} - 477}{1080}. \quad (39)$$

These three cases exhaust every actual cubic zero, because every such zero satisfies the necessary equations (E1)–(E8).

**A.5. Evaluation of the quintic functional.** For the actual root  $gu_{ij} = (p_i - p_j)/\sqrt{2}$ ,

$$|z_1|^2 W(z) = \frac{1}{4\sqrt{2}}(a_i - a_j)^3(\bar{a}_i - \bar{a}_j)(\bar{b}_i - \bar{b}_j). \quad (40)$$

Define

$$\Sigma = \sum_{i < j} X(g)_{ij}(a_i - a_j)^3(\bar{a}_i - \bar{a}_j)(\bar{b}_i - \bar{b}_j). \quad (41)$$

Then

$$\Lambda_g(Y_H(g)) = \frac{1}{4\sqrt{2}} \operatorname{Im} \Sigma. \quad (42)$$

In wedge notation,

$$X(g) = \frac{i}{2}(\bar{a} \wedge a + 2\bar{b} \wedge b). \quad (43)$$

Modulo an exact matrix, the degree-five edge expression in (41) is

$$\begin{aligned} F \equiv & \bar{b} \wedge (a^3 \bar{a}) + \bar{a} \wedge (a^3 \bar{b}) - (\bar{a} \bar{b}) \wedge a^3 \\ & + 3a \wedge (a^2 \bar{a} \bar{b}) - 3(a \bar{b}) \wedge (a^2 \bar{a}) \\ & - 3(a \bar{a}) \wedge (a^2 \bar{b}) + 3(a \bar{a} \bar{b}) \wedge a^2. \end{aligned} \quad (44)$$

The omitted exact term is  $(a^3 \bar{a} \bar{b}) \wedge \mathbf{1}$ , which is annihilated by  $X(g)$ . For the bilinear dot product  $x \cdot y = \sum_i x_i y_i$ ,

$$\sum_{i < j} (u \wedge v)_{ij} (x \wedge y)_{ij} = (u \cdot x)(v \cdot y) - (u \cdot y)(v \cdot x). \quad (45)$$

Every dot product in (45) is five times the coefficient of  $\mathbf{1}$  in the corresponding coordinatewise product. Substituting (27), (43), and (44) gives

$$\Sigma = -\frac{i}{80} \mathcal{B}_{\text{raw}}, \quad (46)$$

where

$$\begin{aligned} \mathcal{B}_{\text{raw}} = & -960\bar{k} \ell t + 160 |\ell|^2 m + 320 \ell t^2 - 120 m^3 \\ & - 75 m |n|^2 - 800 m t^2 + 100 m |v|^2 + 192 m - 160 n t \bar{v}. \end{aligned} \quad (47)$$

In all three cases above,  $\ell$  is real. The conjugates of (E4) and (E1) give

$$\bar{k} t = \ell m + 4t^2, \quad n t \bar{v} = \frac{3}{4} m |n|^2. \quad (48)$$

Using (48), followed by (E6) and (E8), reduces (47) to

$$\mathcal{B} = m(180m^2 + 105 |n|^2 - 4400t^2 - 128) - 3520 \ell t^2. \quad (49)$$

The three cases now give

| case | $\mathcal{B}$           |
|------|-------------------------|
| I    | $160m$                  |
| II   | $-80m$                  |
| III  | $60m(75 - 7\sqrt{105})$ |

(50)

Each value is nonzero: in every case  $m \neq 0$ , and  $75 - 7\sqrt{105} > 0$ . Since  $\mathcal{B}$  is real, (42) and (46) yield

$$\Lambda_g(Y_H(g)) = -\frac{\mathcal{B}}{320\sqrt{2}} \neq 0.$$

This proves Proposition 3.3.

*Remark A.1* (Symbolic verification). The wedge expansion (44), the extraction of (E1)–(E8), the resultants (36)–(37), and the simplification (49) are polynomial identities. An accompanying exact-arithmetic SymPy script checks each of them independently.

## REFERENCES

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